Exercise Guide 3 — Keep the Lights On: Grid Balancing Game

NISO Staff Acculturation Programme · Day 2

At a glance

When	Tuesday 27 October, 1:00–2:15 pm (75 minutes)
Comes right before	Inside the Control Room: Guided Virtual Tour
Who leads	Engr. Dr. Oladeji Ojo, with the L&P team as "grid inspectors"
Group size	The week's teams of 5 to 7
Scored?	Yes — 10 / 7 / 5 / 3 points

What it is. Each team becomes the dispatcher of a small, make-believe grid. Every round is a time of day with a new demand. Teams move tokens to make generation match demand exactly. Surprise events hit them along the way: a power station trips, the sun sets, a line goes out.

Why we do it. The game teaches how a grid *should* work in an ideal world. Participants feel why supply must match demand every moment, why reserves matter, and why some power stations are slow to change. The control room tour that follows then shows what the real Nigerian grid is and what it takes to run it.

How the game works

One token = 10 megawatts (MW). Each team has a grid board with four power stations and a store of spare tokens. Tokens placed on a station are power being produced.

Power station	Spaces on the board	How fast it can change each round	Who controls it
Hydro	8 tokens (80 MW)	Any amount — it is fast	The team
Gas A	12 tokens (120 MW)	Up or down by 3 tokens at most	The team
Gas B	12 tokens (120 MW)	Up or down by 3 tokens at most	The team
Solar	6 tokens (60 MW)	Set by the sun — the facilitator says how much	The facilitator

The goal each round. Total tokens on all four stations must equal the demand for that round. Too little, and the lights go off. Too much, and the grid is unstable.

The five rules

1. Each round, the facilitator announces the time of day, the demand and the solar output.
2. The team has 2 minutes to move tokens between the store and the stations.
3. Gas A and Gas B can move by 3 tokens at most per round. Breaking this rule scores 0 for that round.
4. **Reserve:** keep at least 2 empty spaces on Hydro. That is your quick back-up if something fails. It earns 1 bonus point.

5. **Load shedding** is the last resort. A team may cut the demand by up to 3 tokens. Each token cut costs 1 point, because customers lose power.

Starting position (before the practice round): Hydro 4, Gas A 6, Gas B 6, Solar 0. That is 16 tokens = 160 MW.

What you need and how to prepare

Materials (per team)

- ☐ One A3 grid board: four boxes marked Hydro (8 spaces), Gas A (12), Gas B (12), Solar (6), plus a "Store" box
- ☐ 40 tokens: bottle tops, buttons, beans or paper squares — all the same size
- ☐ One score sheet
- ☐ One printed copy of the five rules

Materials (for the facilitator)

- ☐ The round script (next sections of this guide)
- ☐ A flip chart or slide showing the round table, revealed one round at a time
- ☐ A bell or whistle, and a timer
- ☐ Event cards: "Gas A trips!" and "Line outage!" written large

Before the day

1. Draw or print the grid boards. One per team plus two spares.
2. Count 40 tokens into a bag for each team.
3. Brief the three L&P helpers as grid inspectors. Each one checks the boards of two or three teams after every round, using the scoring rules.
4. Play one round yourself with the helpers, so everyone knows the rules.

Step by step

Time	Step	The facilitator does	The participants do
0–5 min	1. Set the scene	Says: "For the next hour, you are the dispatchers of a small grid. In this game the world is ideal. After the game, the control room will show you the real Nigerian grid."	Sit with their team around the grid board.
5–15 min	2. Explain the board and rules	Walks through the four stations and the five rules. Shows the starting position. Answers questions.	Place the starting tokens: Hydro 4, Gas A 6, Gas B 6. Choose one "dispatcher" to move tokens and one "checker" to count.
15–20 min	3. Practice round	Runs the practice round from the script. No points. Corrects mistakes kindly.	Play the practice round. Ask questions.
20–56 min	4. Six scored rounds	For each round: announces time, demand and solar (and any event). Starts 2 minutes. Rings the bell: "Hands off!" Gives one teaching line from the script while inspectors score.	Discuss quickly. Move tokens. Stop when the bell rings. Tell the inspector their total.
56–60 min	5. Add up the scores	Collects score sheets. Announces 1st, 2nd and 3rd.	Add up their points. Hand in the sheet.
60–73 min	6. Debrief	Asks the debrief questions. Writes key answers on the flip chart. Lists what was "ideal" in the game.	Share what they felt and learnt.

Time	Step	The facilitator does	The participants do
73–75 min	7. Bridge to the tour	Says: "Keep your list of ideal things. In the tour, watch for how real life is different." Hands out the "Ideal vs Reality" cards.	Take a card. Note one question for the control room engineers.

Each round takes about 6 minutes: 1 minute to announce, 2 minutes to play, 2–3 minutes to score and teach.

The round script

Read each round out loud. Reveal one row at a time on the flip chart. "Tokens needed" is what Hydro + Gas A + Gas B must add up to, after solar.

Round	Time of day	Demand	Solar	Tokens needed	Event (read it out)	Teaching line (say it after scoring)
Practice	5:00 am	160 MW	0	16	None	"Supply must equal demand, every single moment."
1	7:00 am	190 MW	1	18	None	"Morning: people wake up and demand rises. Slow stations must be moved early."
2	10:00 am	230 MW	4	19	None	"Solar helps — but only while the sun shines."
3	1:00 pm	250 MW	5	20	**"Gas A trips! Remove all its tokens now."**	"A trip comes without warning. Your reserve saves the grid. No reserve means load shedding."
4	4:00 pm	240 MW	3	21	"Gas A is back, but can only restart at 3 tokens or less."	"A tripped unit cannot return to full power at once."
5	7:00 pm	260 MW	0	26	None	"Evening peak: demand is highest just as solar drops to zero."
6	10:00 pm	210 MW	0	21	**"Line outage! Hydro can send 4 tokens at most this round."**	"Having power is not enough. You must be able to move it to where it is needed."

Notes for the facilitator

- Every round can be balanced if teams plan ahead. Round 3 is only possible for teams that kept Gas B high and Hydro low in Round 2 — that is the lesson of reserve.
- In Round 3 and Round 5, most teams will use all of Hydro, so no reserve bonus is expected. That is fine: reserve is there to be used.
- In Round 6, the reserve bonus is earned if Hydro has 2 tokens or less.

Scoring, problems and debrief

Scoring each round (grid inspectors)

Gap between generation and demand	Points
Exact match	3
1 token off (10 MW)	2

Gap between generation and demand	Points
2 tokens off (20 MW)	1
3 or more tokens off	0 — show a red "Frequency alarm!" card

Then add or take away:

- **+1** if the team kept its reserve (2 empty spaces on Hydro; in Round 6, Hydro at 2 or less).
- **–1** for each token of load shed.
- **0 for the round** if Gas A or Gas B moved by more than 3 tokens.

At the end, rank teams by total game points. Give 10 points to 1st, 7 to 2nd, 5 to 3rd and 3 to every other team.

If something goes wrong

- *Teams are confused after the practice round:* run the practice round again. It is worth the 5 minutes.
- *Running late:* drop Round 2. Keep Rounds 3, 5 and 6 — they carry the main lessons.
- *Inspectors disagree with a team:* the facilitator decides at once. Keep the game moving.

Debrief questions

1. How did it feel when Gas A tripped?
2. Why did the reserve matter?
3. Why can't a gas station just jump to full power?
4. What happened at 7:00 pm, and why?
5. What made this game easier than real life?

For question 5, write the answers on the flip chart as the "ideal world" list. Expect answers like: gas is always available, demand is known exactly, every line works (until the outage), every DisCo takes the power sent to it, and the dispatcher sees everything at once. Keep this list up for the control room tour.